

Anonymous ARR Rebuttal

Anonymous

1 Additional materials

Model	COH	SYN	PHR	VOC	GRA	CON	Avg.
ModernBERT (Ours)	0.65	0.68	0.66	0.68	0.68	0.70	0.67
NLA (Ours)	0.59	0.65	0.49	0.64	0.47	0.64	0.58
RAvg (Ours)	0.62	0.65	0.61	0.62	0.68	0.64	0.64
Q-CFA (Ours)	0.62	0.64	0.63	0.61	0.69	0.65	0.64
Q-MIRT (Ours)	0.62	0.64	0.65	0.62	0.68	0.64	0.64
BERT [1]	0.57	0.66	0.65	0.63	0.63	0.66	0.63
RoBERTa [1]	0.62	0.67	0.69	0.65	0.71	0.71	0.67
MTL _{vanilla} [1]	0.63	0.69	0.72	0.67	0.72	0.71	0.69
MTAA [1]	0.63	0.69	0.71	0.67	0.73	0.72	0.69

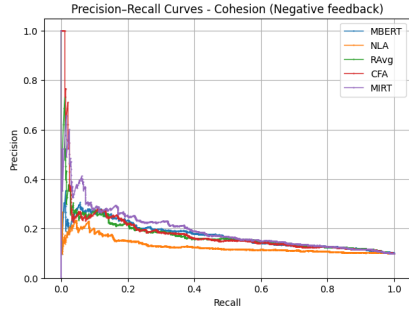
Table 1: Comparison of our ModernBERT-based grader against [1]’s graders in terms of Spearman’s rank coefficient against the original ELLIPSE test set labels.

LLM	Model	Analytic Dimension				
		CON	ORG	LAN	VOC	MCH
Qwen 2.5 72B	NLA	21.46	16.34	14.23	20.87	22.66
	Q-RAvg	22.01	27.51	15.37	16.31	25.87
	Q-WAvg	23.36	29.77	14.90	14.86	29.47
	Q-CFA	21.94	24.27	15.35	14.52	17.84
	Q-MIRT	20.90	28.44	16.11	18.20	19.12
gpt-4.1-mini	NLA	22.79	25.26	17.40	12.87	17.04
	Q-RAvg	28.16	37.15	16.31	16.80	23.31
	Q-WAvg	25.62	34.10	16.80	16.67	24.68
	Q-CFA	25.42	31.98	16.04	19.26	19.80
	Q-MIRT	23.11	33.96	15.76	18.75	18.67

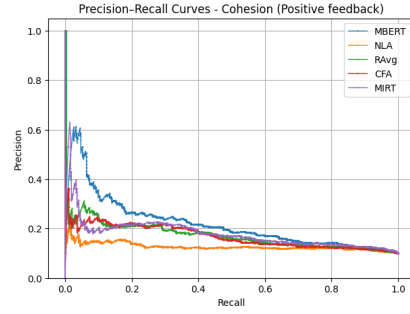
Table 2: Self-referential evaluation in terms of $F_{0.5}$ for **Negative** feedback (IC-NALE EE). **Qwen 2.5 72B** vs **gpt-4.1-mini**.

LLM	Model	Analytic Dimension				
		CON	ORG	LAN	VOC	MCH
Qwen 2.5 72B	NLA	14.15	27.23	24.26	12.32	18.82
	Q-RAvg	17.36	20.41	26.59	22.46	19.94
	Q-WAvg	16.31	21.22	26.16	19.21	19.07
	Q-CFA	14.17	20.64	24.75	18.03	18.67
	Q-MIRT	19.93	19.43	23.52	19.86	14.86
gpt-4.1-mini	NLA	17.55	20.60	17.99	15.86	29.02
	Q-RAvg	17.38	21.16	27.25	29.34	24.68
	Q-WAvg	17.87	22.21	26.98	27.68	27.60
	Q-CFA	21.13	19.62	27.14	31.34	22.14
	Q-MIRT	20.09	20.67	28.79	26.98	18.77

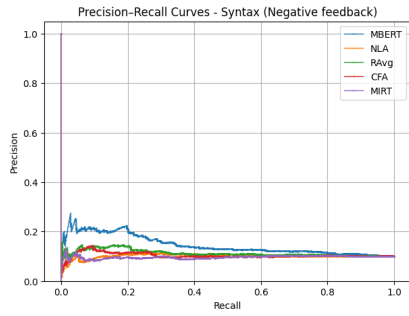
Table 3: Self-referential evaluation in terms of $F_{0.5}$ for **Positive** feedback (IC-NALE EE), **Qwen 2.5 72B** vs **gpt-4.1-mini**.



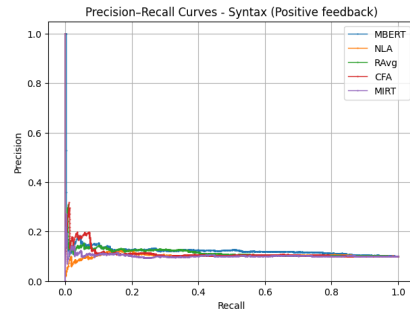
(a) Cohesion - negative feedback



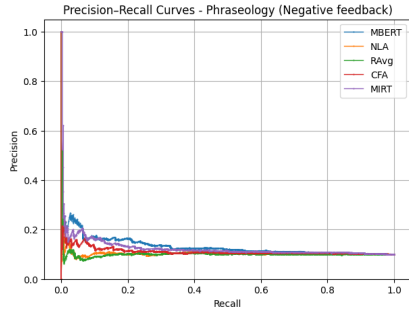
(b) Cohesion - positive feedback



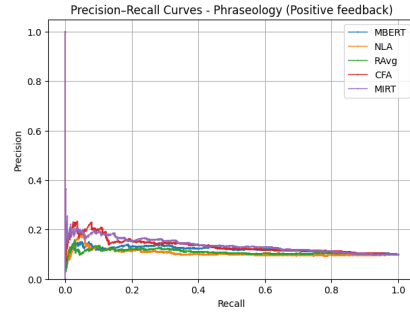
(c) Syntax - negative feedback



(d) Syntax - positive feedback

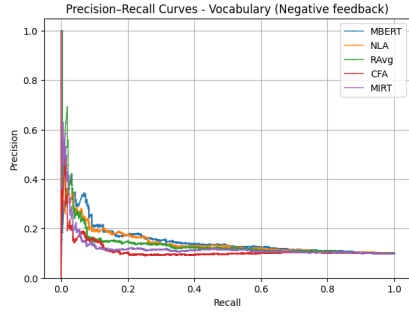


(e) Phraseology - negative feedback

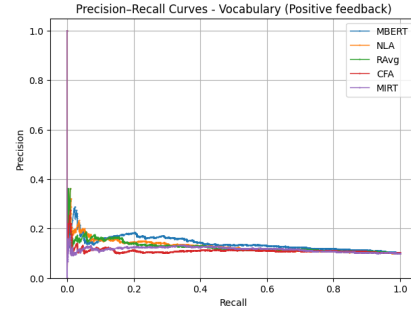


(f) Phraseology - positive feedback

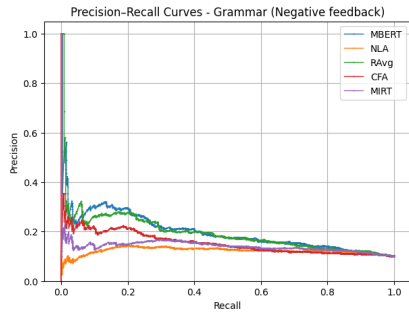
Figure 1: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (1 of 2).



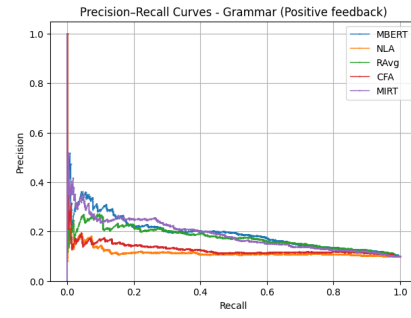
(a) Vocabulary - negative feedback



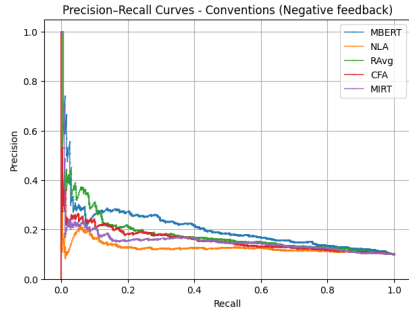
(b) Vocabulary - positive feedback



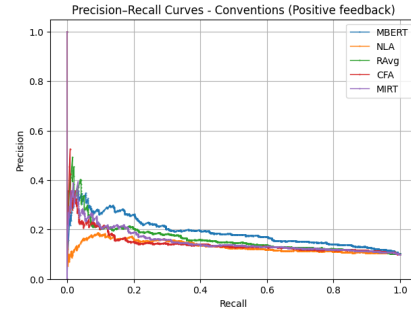
(c) Grammar - negative feedback



(d) Grammar - positive feedback



(e) Conventions - negative feedback



(f) Conventions - positive feedback

Figure 2: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (2 of 2).

References

- [1] S. Chen, Y. Lan, and Z. Yuan, “A multi-task automated assessment system for essay scoring,” in *International Conference on Artificial Intelligence in Education*, pp. 276–283, Springer, 2024.